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Methods for detection and mitigation of well screen out

Abstract

Methods, systems, and controllers for detecting and mitigating well screen outs may include a controller configured to operate a fracturing pump to supply fluid at a discharge rate to a wellhead at a fracturing well site. The controller may also operate a blender positioned to deliver a blend of proppant and fluid to the fracturing pump. The controller may compare a fluid pressure increase rate to a preselected increase rate indicative of a potential well screen out. The controller may incrementally decrease the discharge rate of the fracturing pump and a flow rate of a blender when the fluid pressure increase rate of the wellhead exceeds the preselected increase rate and the fluid pressure is within a preselected percentage of a maximum wellhead pressure until the fluid pressure of the fluid supplied to the wellhead is stabilized.

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10337402	12/2018	Austin et al.	N/A	N/A
10358035	12/2018	Cryer et al.	N/A	N/A
10371012	12/2018	Davis et al.	N/A	N/A
10374485	12/2018	Morris et al.	N/A	N/A
10378326	12/2018	Morris et al.	N/A	N/A
10393108	12/2018	Chong et al.	N/A	N/A
10407990	12/2018	Oehring et al.	N/A	N/A
10408031	12/2018	Oehring et al.	N/A	N/A
10415348	12/2018	Zhang et al.	N/A	N/A
10415557	12/2018	Crowe et al.	N/A	N/A
10415562	12/2018	Kajita et al.	N/A	N/A
10422207	12/2018	Aidagulov et al.	N/A	N/A
RE47695	12/2018	Case et al.	N/A	N/A
10465689	12/2018	Crom	N/A	N/A
10478753	12/2018	Elms et al.	N/A	N/A
10526882	12/2019	Oehring et al.	N/A	N/A
10563649	12/2019	Zhang et al.	N/A	N/A
10570704	12/2019	Colvin et al.	N/A	N/A
10577908	12/2019	Kisra et al.	N/A	N/A
10577910	12/2019	Stephenson	N/A	N/A
10584645	12/2019	Nakagawa et al.	N/A	N/A
10590867	12/2019	Thomassin et al.	N/A	N/A
10598258	12/2019	Oehring et al.	N/A	N/A
10605060	12/2019	Chuprakov et al.	N/A	N/A
10610842	12/2019	Chong	N/A	N/A
10662749	12/2019	Hill et al.	N/A	N/A
10677961	12/2019	Chen et al.	N/A	N/A
10711787	12/2019	Darley	N/A	N/A
10738580	12/2019	Fischer et al.	N/A	N/A
10753153	12/2019	Fischer et al.	N/A	N/A
10753165	12/2019	Fischer et al.	N/A	N/A
10760416	12/2019	Weng et al.	N/A	N/A
10760556	12/2019	Crom et al.	N/A	N/A
10794165	12/2019	Fischer et al.	N/A	N/A
10794166	12/2019	Reckels et al.	N/A	N/A
10801311	12/2019	Cui et al.	N/A	N/A
10815764	12/2019	Yeung et al.	N/A	N/A
10815978	12/2019	Glass	N/A	N/A
10830032	12/2019	Zhang et al.	N/A	N/A
10830225	12/2019	Repaci	N/A	N/A
10851633	12/2019	Harper	N/A	N/A
10859203	12/2019	Cui et al.	N/A	N/A
10864487	12/2019	Han et al.	N/A	N/A
10865624	12/2019	Cui et al.	N/A	N/A
10865631	12/2019	Zhang et al.	N/A	N/A
10870093	12/2019	Zhong et al.	N/A	N/A
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10895202	12/2020	Yeung et al.	N/A	N/A
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10954770	12/2020	Yeung et al.	N/A	N/A
10954855	12/2020	Ji et al.	N/A	N/A
10961614	12/2020	Djavanroodi	N/A	N/A
10961908	12/2020	Yeung et al.	N/A	N/A
10961912	12/2020	Yeung et al.	N/A	N/A

10961914	12/2020	Yeung et al.	N/A	N/A
10961993	12/2020	Ji et al.	N/A	N/A
10961995	12/2020	Mayorca	N/A	N/A
10968837	12/2020	Yeung et al.	N/A	N/A
10982523	12/2020	Hill et al.	N/A	N/A
10989019	12/2020	Cai et al.	N/A	N/A
10989180	12/2020	Yeung et al.	N/A	N/A
10995564	12/2020	Miller et al.	N/A	N/A
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11015423	12/2020	Yeung et al.	N/A	N/A
11015536	12/2020	Yeung et al.	N/A	N/A
11015594	12/2020	Yeung et al.	N/A	N/A
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11035213	12/2020	Dusterhoft et al.	N/A	N/A
11035214	12/2020	Cui et al.	N/A	N/A
11047379	12/2020	Li et al.	N/A	N/A
11053853	12/2020	Li et al.	N/A	N/A
11060455	12/2020	Yeung et al.	N/A	N/A
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11068455	12/2020	Shabi et al.	N/A	N/A
11085281	12/2020	Yeung et al.	N/A	N/A
11085282	12/2020	Mazrooe et al.	N/A	N/A
11092152	12/2020	Yeung et al.	N/A	N/A
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11105250	12/2020	Zhang et al.	N/A	N/A
11105266	12/2020	Zhou et al.	N/A	N/A
11109508	12/2020	Yeung et al.	N/A	N/A
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11125156	12/2020	Zhang et al.	N/A	N/A
11129295	12/2020	Yeung et al.	N/A	N/A
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11156159	12/2020	Yeung et al.	N/A	N/A
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11174716	12/2020	Yeung et al.	N/A	N/A
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11205880	12/2020	Zhou	N/A	N/A
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11242737	12/2021	Zhang et al.	N/A	N/A
11243509	12/2021	Cai et al.	N/A	N/A
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11261717	12/2021	Yeung et al.	N/A	N/A
11268346	12/2021	Yeung et al.	N/A	N/A
11280266	12/2021	Yeung et al.	N/A	N/A
11306835	12/2021	Dille et al.	N/A	N/A
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11319791	12/2021	Yeung	N/A	E21B 21/08
11339638	12/2021	Yeung et al.	N/A	N/A
11346200	12/2021	Cai et al.	N/A	N/A
11373058	12/2021	Jaaskelainen et al.	N/A	N/A
RE49140	12/2021	Case et al.	N/A	N/A
11377943	12/2021	Kriebel et al.	N/A	N/A
RE49155	12/2021	Case et al.	N/A	N/A
RE49156	12/2021	Case et al.	N/A	N/A
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11448122	12/2021	Feng et al.	N/A	N/A
11466680	12/2021	Yeung et al.	N/A	N/A
11480040	12/2021	Han et al.	N/A	N/A
11492887	12/2021	Cui et al.	N/A	N/A
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11512570	12/2021	Yeung et al.	N/A	N/A
11519395	12/2021	Zhang et al.	N/A	N/A
11519405	12/2021	Deng et al.	N/A	N/A
11530602	12/2021	Yeung et al.	N/A	N/A
11549349	12/2022	Wang et al.	N/A	N/A
11555390	12/2022	Cui et al.	N/A	N/A
11555756	12/2022	Yeung et al.	N/A	N/A
11557887	12/2022	Ji et al.	N/A	N/A
11560779	12/2022	Mao et al.	N/A	N/A
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11575249	12/2022	Ji et al.	N/A	N/A
11592020	12/2022	Chang et al.	N/A	N/A
11596047	12/2022	Liu et al.	N/A	N/A
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11603797	12/2022	Zhang et al.	N/A	N/A
11607982	12/2022	Tian et al.	N/A	N/A
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11649819	12/2022	Gillispie	N/A	N/A
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11668289	12/2022	Chang et al.	N/A	N/A
11677238	12/2022	Liu et al.	N/A	N/A
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2006/0211356	12/2005	Grassman	N/A	N/A
2006/0228225	12/2005	Rogers	N/A	N/A
2006/0260331	12/2005	Andreychuk	N/A	N/A
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2007/0107981	12/2006	Sicotte	N/A	N/A
2007/0125544	12/2006	Robinson et al.	N/A	N/A
2007/0169543	12/2006	Fazekas	N/A	N/A

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2007/0272407	12/2006	Lehman et al.	N/A	N/A
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2008/0161974	12/2007	Alston	N/A	N/A
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2009/0178412	12/2008	Spytek	N/A	N/A
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2016/0273328	12/2015	Oehring	N/A	N/A
2016/0273346	12/2015	Tang et al.	N/A	N/A
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2016/0326845	12/2015	Djikpesse et al.	N/A	N/A
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2017/0009905	12/2016	Arnold	N/A	N/A
2017/0016433	12/2016	Chong et al.	N/A	N/A
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2017/0038137	12/2016	Turney	N/A	N/A
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2017/0051598	12/2016	Ouenes	N/A	N/A
2017/0052087	12/2016	Faqihi et al.	N/A	N/A
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2017/0114613	12/2016	Lecerf et al.	N/A	N/A
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2017/0131174	12/2016	Enev et al.	N/A	N/A
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2017/0218727	12/2016	Oehring et al.	N/A	N/A
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2017/0234165	12/2016	Kersey et al.	N/A	N/A
2017/0234308	12/2016	Buckley	N/A	N/A
2017/0241336	12/2016	Jones et al.	N/A	N/A
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2017/0247995	12/2016	Crews et al.	N/A	N/A
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2017/0306936	12/2016	Dole et al.	N/A	N/A
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2018/0229998	12/2017	Shock	N/A	N/A
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2019/0003272	12/2018	Morris et al.	N/A	N/A
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2020/0072201	12/2019	Marica	N/A	N/A
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Background/Summary

PRIORITY CLAIM (1) This is a continuation of U.S. Non-Provisional application Ser. No. 17/991,007, filed Nov. 21, 2022, titled "METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT," now U.S. Pat. No. 11,939,854 issued Mar. 26, 2024, which is a continuation of U.S. Non-Provisional application Ser. No. 17/355,920, filed Jun. 23, 2021, titled "METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT," now U.S. Pat. No. 11,566,506 issued Jan. 31, 2023, which is continuation of U.S. Non-Provisional application Ser. No. 17/303,841, filed Jun. 9, 2021, titled "METHODS AND SYSTEMS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT," now U.S. Pat. No. 11,208,881, issued Dec. 28, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 17/182,408, filed Feb. 23, 2021, titled "METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT," now U.S. Pat. No. 11,066,915, issued Jul. 20, 2021, which claims priority to and the benefit of U.S. Provisional Application No. 62/705,050, filed Jun. 9, 2020, titled "METHODS AND SYSTEMS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT," the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The application generally relates to mobile power units and, more specifically, drive equipment and methods for usage, installation on, and controls for mobile fracturing transportation platforms.

BACKGROUND

(2) Hydrocarbon exploration and energy industries employ various systems and operations to accomplish activities including drilling, formation evaluation, stimulation and production. Measurements such as temperature, pressure, and flow measurements are typically performed to monitor and assess such operations. During such operations, problems or situations may arise that may have a detrimental effect on the operation, equipment, and/or safety of operators. For example, during a stimulation or fracturing operation, screen out conditions may occur, which may cause rapid pressure increases that may compromise the operation and/or damage equipment.

SUMMARY

(3) Embodiment of systems, methods, and controllers that control the operation to detect and mitigate screen outs such that screen

outs are avoided, for example, may save time, may increase awareness of conditions within the well, and may increase safety at a wellsite hydraulic fracturing pumper system. For example, Applicant has recognized that a controller detecting and mitigating screen outs may avoid packing of a well and avoid the need for additional operations to stimulate a well, e.g., wire line operations. In addition, a controller that avoids rapid pressure increases associated with screen outs may reduce stress on fracturing equipment including power end assemblies, shocking of prime movers and gearing systems associated therewith, and piping of the well. Further, the methods and systems detailed herein may prevent energy release in the form of release pressure through a pressure relief valve, e.g., a wellhead or manifold pressure relief valve. Avoiding pressure release from a pressure valve may also increase the safety of the wellhead, for example, by not over pressuring a wellhead.

(4) Applicant also has recognized that a controller that detects and mitigates screen outs may also increase awareness of conditions within the well by detecting a rate of pressure increase more accurately and at a more frequent rate than with manual control. In some embodiments, the controller may prewarn by one or more tiers of pressure increase rates such that an operator may manually adjust proppant concentration or take other measures to avoid screen outs before the controller intervenes as would be appreciated by those skilled in the art. The controller may also control the blender and the fracturing pump with a single command such that an operator is not required to sequence both elements in a safe manner to avoid damage to equipment, e.g., via cavitation, and to avoid screen out.

(5) In accordance with an embodiment of the present disclosure, a method of detecting and mitigating well screen out at a fracturing well site during hydrocarbon production may include operating a fracturing pump to supply fluid at a discharge rate to a wellhead at a fracturing well site. The method also may include operating a blender positioned to deliver a blend of proppant and fluid to the fracturing pump. A fluid pressure of the fluid supplied to the wellhead may be measured and a fluid pressure increase rate of the fluid may be determined from the fluid pressure. The fluid pressure increase rate may be compared to a preselected increase rate indicative of a potential well screen out. When the fluid pressure increase rate exceeds the preselected increase rate and the fluid pressure is within a preselected percentage of a maximum wellhead pressure of the well head, the discharge rate of the fracturing pumps may be incrementally decreased until the fluid pressure increase rate is stabilized. Stabilizing the fluid pressure increase rate may include the fluid pressure increase rate being equal to or less than zero.

(6) In accordance with another embodiment of the present disclosure, a wellsite hydraulic fracturing pumper system may include one or more fracturing pumps, a blender, a pressure transducer, and a controller. The one or more fracturing pumps may be configured to provide fluid to a wellhead when positioned a hydrocarbon well site. The blender may be configured to provide fluid and proppant to the one or more fracturing pumps. The pressure transducer may be positioned adjacent an output of the one or more fracturing pumps or at the wellhead. The pressure transducer may be configured to measure a fluid pressure of the fluid provided to the wellhead. The controller may control the one or more fracturing pumps and the blender. The controller may be positioned in signal communication with the pressure transducer such that the controller receives the fluid pressure of the fluid provided to the wellhead. The controller may include memory, a processor to process data, and a screen out detection and mitigation protocol program stored in the memory and responsive to the process and in which the protocol of the controller may incrementally decrease a discharge rate of the one or more fracturing pumps and a flow rate of the blender in response to a fluid pressure increase rate of the fluid supplied to the wellhead being greater than a preselected increase rate and the fluid pressure of the fluid provided to the wellhead being greater than a preselected percentage of a maximum wellhead pressure until the fluid pressure is stabilized.

(7) In yet another embodiment of the present disclosure, a controller for a hydraulic fracturing pumper system may include a pressure input, a first control output, and a second control output. The pressure input may be in signal communication with a pressure transducer that measures a fluid pressure of a fluid being provided to a wellhead. The first control output may be in signal communication with a fracturing pump such that the controller provides pump control signals to the fracturing pump to control a discharge rate of the fracturing pump. The second control output may be in signal communication with a blender such that the controller provides blender control signals to the blender to control a flow rate of the blender and delivery of a proppant from the blender. The controller may be configured to calculate a fluid pressure increase rate of the fluid pressure, compare the fluid pressure increase rate of the fluid pressure to a preselected increase rate, and incrementally decrease a discharge rate of the fracturing pump and a flow rate of the blender when the fluid pressure increase rate is greater than the preselected increase rate and the fluid pressure is within a preselected percentage of a maximum wellhead pressure of the wellhead until the fluid pressure of the fluid is supplied to the wellhead is stabilized.

(8) Those skilled in the art will appreciate the benefits of various additional embodiments reading the following detailed description of the embodiments with reference to the below-listed drawing figures. It is within the scope of the present disclosure that the above-discussed embodiments and aspects be provided both individually and in various combinations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the embodiments of the present disclosure, are incorporated in and constitute a part of this specification, and together with the detailed description, serve to explain the principles of the embodiments discussed herein. The present disclosure may be more readily described with reference to the accompanying drawings.

(2) FIG. 1 is a schematic view of a wellsite hydraulic fracturing pumper system according to an embodiment of the disclosure.

(3) FIG. 2 is a schematic view of a control system of the wellsite hydraulic fracturing pumper system of FIG. 1.

(4) FIG. 3 is a flowchart of a method of detecting and mitigating a well screen out of a well according to an embodiment of the present disclosure.

(5) Corresponding parts are designated by corresponding reference numbers throughout the drawings.

DETAILED DESCRIPTION

(6) The present disclosure will now be described more fully hereinafter with reference to example embodiments thereof with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. These example embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Features from one embodiment or aspect may be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments may be applied to apparatus, product, or component aspects or embodiments and vice versa. The disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification and the appended claims, the singular forms “a,” “an,” “the,” and the like include plural referents unless the context clearly dictates otherwise. In addition, while reference may be made herein to quantitative measures, values, geometric relationships or the like, unless otherwise stated, any one or more if not all of these may be absolute or approximate to account for acceptable variations that may occur, such as those due to manufacturing or engineering tolerances or the like.

(7) Embodiments of the present disclosure are directed to methods and systems for detecting and mitigating well screen outs during the operations of wellsite hydraulic fracturing pumping systems during the production of hydrocarbons. The methods and systems detailed herein may be executed on a controller that provides alerts or alarms to an operator of a potential well screen out and may intervene to prevent the fluid pressure provided to the well from exceeding a maximum well pressure.

(8) FIG. 1 illustrates an exemplary wellsite hydraulic fracturing pumper system **1000** that is provided in accordance with an embodiment of the present disclosure. The wellsite hydraulic fracturing pumper system **1000** includes a plurality of mobile power units **100** arranged around a wellhead **10** to supply the wellhead **10** with high-pressure fracturing fluids and recover oil and/or gas from the wellhead **10** as will be understood by those skilled in the art. As shown, some of the mobile power units **100**, e.g., mobile power units **100a**, drive a hydraulic fracturing pump **200** that discharges high pressure fluid to a manifold **20** such that the high pressure fluid is provided to the wellhead **10**. Additionally, some of the mobile power units **100**, e.g., mobile power units **100b**, drive an electrical generator **300** that provides electrical power to the wellsite hydraulic fracturing pumper system **1000**.

(9) The wellsite hydraulic fracturing pumper system **1000** also includes a blender unit **410**, a hydration unit **420**, or a chemical additive unit **430** which may be referred to generally as backside equipment **400**. Specifically, the blender unit **410** provides a flow of fluid to the fracturing pumps **200** which is pressurized by and discharged from the fracturing pumps **200** into the manifold **20**. The blender unit **410** may include one or more screw conveyors **412** that provides proppant to a mixer **416** of the blender unit **410**. The blender unit **410** also includes a discharge pump **418** that draws fluid from the mixer **416** such that a flow of fluid is provided from the blender unit **410** to the fracturing pumps **200**. The fluid from the mixer **416** may include proppant provided by the screw conveyors **412** and/or chemicals for the fluid of the fracturing pumps **200**. When blender unit **410** provides proppant to the fracturing pumps **200**, the proppant is in a slurry which may be considered a fluid as will be understood by those skilled in the art.

(10) The wellsite hydraulic fracturing pumper system **1000** includes a supervisory control unit that monitors and controls operation of the mobile power units **100a** driving the fracturing pumps **200**, the mobile power units **100b** driving electrical generators **300**, and the units **410**, **420**, **430** and may be referred to generally as controller **30**. The controller **30** may be a mobile control unit in the form of a trailer or a van, as appreciated by those skilled in the art. As used herein, the term “fracturing pump” may be used to refer to one or more of the hydraulic fracturing pumps **200** of the hydraulic fracturing pumper system **1000**. In some embodiments, all of the hydraulic fracturing pumps **200** are controlled by the controller **30** such that to an operator of the controller **30**, the hydraulic fracturing pumps **200** are controlled as a single pump or pumping system.

(11) The controller **30** is in signal communication with the blender unit **410** to control the delivery of the proppant to the mixer **416** and a flow rate of fluid from the discharge pump **418** to the fracturing pumps **200**. The controller **30** is also in signal communication with the fracturing pumps **200** to control a discharge rate of fluid from the fracturing pumps **200** into the manifold **20**. In addition, the controller **30** is in signal communication with one or more sensors of the wellsite hydraulic fracturing pumper system **1000** to receive measurements or data with respect to the fracturing operation. For example, the controller **30** receives a measurement of pressure of the fluid being delivered to the wellhead **10** from a wellhead pressure transducer **13**, a manifold pressure transducer **23**, or a pump output pressure transducer **213**. The wellhead pressure transducer **13** is disposed at the wellhead **10** to measure a pressure of the fluid at the wellhead **10**. The manifold pressure transducer **23** is shown at an end of the manifold **20**. However, as understood by those skilled in the art, the pressure within the manifold **20** is substantially the same throughout the entire manifold **20** such that the manifold pressure transducer **23** may be disposed anywhere within the manifold **20** to provide a pressure of the fluid being delivered to the wellhead **10**. The pump output pressure transducer **213** is disposed adjacent an output of one of the fracturing pumps **200** which is in fluid communication with the manifold **20** and thus, the fluid at the output of the fracturing pumps **200** is at substantially the same pressure as the fluid in the manifold **20** and the fluid being provided to the wellhead **10**. Each of the fracturing pumps **200** may include a pump output pressure transducer **213** and the controller **30** may calculate the fluid pressure provided to the wellhead **10** as an average of the fluid pressure measured by each of the pump output pressure transducers **213**.

(12) The controller **30** is also in signal communication with sensors disposed about the blender unit **410**. For example, the blender unit **410** may include a blender screw encoder/pickup **411** that provides a rotation rate of the screw conveyors **412** of the blender unit **410** which provide proppant to the mixer **416** such that proppant is provided to the fracturing pumps **200**. When the screw conveyors **412** are not active or rotating, proppant is not being added to the mixer **416** such that no proppant is being provided to the fracturing pumps **200**. The blender unit **410** may include a blender flow meter **413** that measures a flow of fluid from the blender unit **410** to the fracturing pumps **200**.

(13) As used herein, “signal communication” refers to electric communication such as hard wiring two components together or wireless communication, as understood by those skilled in the art. For example, wireless communication may be Wi-Fi®, Bluetooth®, ZigBee, or forms of near field communications. In addition, signal communication may include one or more intermediate controllers or relays disposed between elements that are in signal communication with one another. For example, a pump output pressure transducer **213** may be in direct electrical communication with a pump controller (not explicitly shown) and the pump controller may be in direct electrical communication or wireless communication with a master controller (not explicitly

shown) of the mobile power unit **100** which is in electrical or wireless communication with the controller **30**.

(14) FIG. 2 illustrates a schematic of a control system for the wellsite hydraulic fracturing pumper system **1000** referred to generally as a control system **1010**. The control system **1010** includes the controller **30** that is in signal communication with the wellhead pressure transducer **13**, a manifold pressure transducer **23**, and a pump output transducer **213**. The controller **30** includes memory **32** and a processor **34**. The memory **32** may be loaded or preloaded with programs, e.g., detection and mitigation protocol programs as detailed below, that are executed on the processor **34**. The pump output transducer **213** may be in direct signal or electrical communication with a pump controller **215** which may be in direct signal or electrical communication with a mobile power unit controller **105** with the mobile power unit controller **105** in direct signal or electrical communication with the controller **30** such that the pump output transducer **213** is in signal communication with the controller **30**. In some embodiments, the pump output transducer is in direct signal communication with the controller **30**. The pump controller **215** is configured to control the fracturing pump **200** in response to commands signals provided by the controller **30** or the mobile power unit controller **30**. The pump controller **215** may include a pump profiler that records events experienced by the fracturing pump **200**. The recorded events may be used to schedule maintenance of the fracturing pump **200**.

(15) The control system **1010** may include a blender controller **419**, a blender flow meter **413**, and a blender screw encoder/pickup **411**. The blender flow meter **413** and the blender screw encoder/pickup **411** may be in direct signal or electrical communication with the blender controller **419** which may be in direct signal or electrical communication with the controller **30** such that the blender flow meter **413** and the blender screw encoder/pickup **411** are in signal communication with the controller **30**.

(16) FIG. 3 illustrates a method of detecting and mitigating well screen out for a hydraulic fracturing operation is described in accordance with embodiments of the present disclosure and is referred to generally as method **500**. The method **500** is detailed with reference to the wellsite hydraulic fracturing pumper system **1000** and the control system **1010** of FIGS. 1 and 2. Unless otherwise specified, the actions of the method **500** may be completed within the controller **30**. Specifically, the method **500** may be included in one or more programs or protocols loaded into the memory **32** of the controller **30** and executed on the processor **34**. The well screen out protocol is activated (Step **501**) either automatically when the controller **30** is started or may be manually activated by an operator. When well screen out protocol is activated, a maximum wellhead pressure is provided to the controller **30** (Step **510**). The maximum wellhead pressure may be input by an operator into a human interface of the controller **30** or may be a preselected pressure programmed into the controller **30**. When the maximum wellhead pressure is provided by an operator, the controller may verify that the inputted maximum wellhead pressure is within a preselected range. If the inputted maximum wellhead pressure is outside of the preselected range, the controller **30** may display an alarm or reject the inputted maximum wellhead pressure and request another value be inputted by the operator and verify the new inputted maximum wellhead pressure until the inputted maximum wellhead pressure is within the preselected range. The preselected range may be in a range of up to 15,000 per square inch (psi), for example, as will be understood by those skilled in the art.

(17) With the maximum wellhead pressure, the controller **30** verifies that the wellsite hydraulic fracturing pumper system **1000** is in a pumping mode in which at least one of the fracturing pumps **200** is active and that the blender unit **410** is adding proppant to the fluid provided to the fracturing pumps **200** (Step **520**). The controller **30** may verify the blender unit **410** is adding proppant from verifying that one or more of the screw conveyors **412** is rotating via the blender screw encoder/pickups **411**. If either the wellsite hydraulic fracturing pumper system **1000** is not in a pumping mode or that the blender unit **410** is not adding proppant to the fluid being supplied to the fracturing pumps **200** the method **500** is terminated or deactivated. The method **500** may be reactivated manually or when the fracturing pumps **200** and the blender unit **410** are activated to provide fluid including proppant to the wellhead **10**.

(18) Continuing to refer to FIG. 3, when the fracturing pumps **200** and the blender unit **410** are activated to provide fluid including proppant to the wellhead **10**, the controller **30** monitors a fluid pressure of fluid being provided to the wellhead **10** to detect a potential screen out within the well (Step **530**). The fluid pressure of the fluid provided to the wellhead **10** may be monitored from the wellhead pressure transducer **13**, the manifold pressure transducer **23**, the pump output pressure transducers **213**, or combinations thereof. To detect for a potential screen out within the well, the controller **30** monitors a rate of increase of the fluid pressure of fluid being provided to the wellhead **10** which is referred to as fluid pressure increase rate. The fluid pressure increase rate may be calculated by comparing the fluid pressure at a first time $P(t_{sub.1})$ and fluid pressure at a second time $P(t_{sub.2})$ such that the fluid pressure increase rate is calculated as:

$$(19) \text{FluidPressureIncreaseRate} = \frac{P}{t} = \frac{P(t_2) - P(t_1)}{t_2 - t_1}.$$

The fluid pressure may be sampled at a rate in a range of 1 Hertz (Hz) to 300 Hz and the fluid pressure increase rate may be smoothed by taking an average of 2 samples to 100 samples to prevent a single spike of a sample or an erroneous sample from triggering the detection of a potential screen out.

(20) The calculated fluid pressure increase rate is compared to a preselected increase rate to determine if there is a potential for screen out within the well (Step **540**). The preselected increase rate may be an increase rate that is entered by an operator or may be preprogrammed into the controller **30**. The preselected increase rate may be based on historical data of well screen out from other wells, for example, or specific to the well being monitored, as will be understood by those skilled in the art. When the fluid pressure increase rate is below the preselected increase rate, the controller **30** continues to monitor the fluid pressure increase rate while proppant is being added to the fluid provided to the fracturing pumps **200**.

(21) When the fluid pressure increase rate meets or exceeds the preselected increase rate, a tier of the fluid pressure increase rate may be determined (Step **542**). For example, when the fluid pressure increase rate is in a first range of 600 psi/s to 800 psi/s such that the fluid pressure increase rate is a Tier 1 Potential Screen Out and the potential for screen out may be minor. When the fluid pressure increase rate is a Tier 1 Potential Screen Out, the controller **30** provides an alert or message to an operator that the fluid pressure increase rate is high or there is a potential for screen out (Step **544**). The message or alert may be a warning light, a message on a screen, an audible alert, or combinations thereof. In response to the alert or message, an operator may take no action, reduce or stop the addition of proppant to the fluid provided to the fracturing pumps **200**, or reduce a discharge rate of the fracturing pumps **200**.

(22) Continuing with the example, when the fluid pressure increase rate is in a second range of 800 psi/s to 1200 psi/s such that the

fluid pressure increase rate is a Tier 2 Potential Screen Out and the potential for screen out is high. When the fluid pressure increase rate is a Tier 2 Potential Screen Out, the controller **30** provides an alarm or message to an operator that the fluid pressure increase rate is high or potential screen out is high (Step **546**). The message or alarm may be a warning light, a message on a screen, an audible alert, or combinations thereof and is escalated from the message or alert provided for a Tier 1 Potential Screen Out. In response to the alarm or message, an operator may take no action, reduce or stop the addition of proppant to the fluid provided to the fracturing pumps **200**, or reduce a discharge rate of the fracturing pumps **200**.

(23) When the fluid pressure increase rate is above the second range, e.g., 1200 psi/s, the potential for screen out is extremely high such that the fluid pressure increase rate is a Tier 3 Potential Screen Out and a screen out is likely. When the fluid pressure increase rate is a Tier 3 Potential Screen Out, a screen out is likely and the controller **30** enters an intervention or mitigation mode to prevent screen out and prevent or reduce damage to the well and the wellsite hydraulic fracturing pumper system **1000** by the mitigation process **550**. When the controller **30** begins the mitigation process **550**, the controller **30** provides an alert or message to an operator that the mitigation process **550** is running. The message or alert may be a warning light, a message on a screen, an audible alert, or combinations thereof and is escalated from the message or alert provided for a Tier 2 Potential Screen Out.

(24) In the mitigation mode, the controller **30** compares the fluid pressure to the maximum wellhead pressure (Step **552**). When the fluid pressure is greater than a first preselected percentage of the maximum wellhead pressure, e.g., 90%, the controller **30** verifies that the blender screw conveyors **412** are not providing proppant to the blender unit **410**, e.g., that the blender screw conveyors **412** are not rotating. If the blender screw conveyors **412** are providing proppant to the blender unit **410**, the controller **30** stops the blender screw conveyors **412** to stop delivery of proppant (Step **554**). When the delivery of proppant is stopped or verified to be stopped, the controller **30** begins to incrementally decrease a discharge rate of the fracturing pumps **200** as defined by process **560**.

(25) The process **560** may include multiple iterations of decreases in a discharge rate of the fracturing pumps **200** by a preselected increment (Step **562**) and determining the fluid pressure increase rate (Step **564**). The process **560** continues to iterate through Steps **562** and **564** until the fluid pressure increase rate is no longer increasing or stabilized, e.g., less than or equal to zero. The preselected increment may be in a range of 0.5 barrels per minute (BPM) to 10 BPM, e.g., 2 BPM. In some embodiments, the preselected increment is less than 5 BPM. The process **560** may include decreasing the discharge rate of the fracturing pumps **200** by the preselected increment (Step **562**) and delaying the determining the fluid pressure increase rate (Step **564**) for a period of time or a number of cycles of the fracturing pump **200**, e.g., 1 second or 25 cycles or revolutions of the fracturing pump **200**. The delay in determining the fluid pressure increase rate may allow for the fluid pressure to react to the decreased discharge rate before the fluid pressure increase rate is determined. During each iteration of the process **560**, the controller **30** may sequence the flow rate of the blender unit **410** and the discharge rate of the fracturing pump **200**. Specifically, the controller **30** may first send a control signal to the fracturing pump **200** to decrease a discharge rate of the fracturing pump **200** by the increment and then send a control signal to the blender unit **410**, e.g., the discharge pump **418** of the blender unit **410**, to decrease a flow rate of fluid to the fracturing pump **200**. By sequencing the blender unit **410** and the fracturing pumps **200** cavitation at the fracturing pumps **200** may be avoided. In addition, by the controller **30** sequencing the blender unit **410** and the fracturing pumps **200**, the need for an operator to manually sequence the blender unit **410** and the fracturing pumps **200** to maintain a safe operation state is removed.

(26) When the fluid pressure increase rate is stabilized such that the fluid pressure is not increasing or is decreasing (e.g., equal to or less than zero), the controller **30** terminates the mitigation process **550** and maintains the discharge rate of the fracturing pumps **200** (Step **570**). When the mitigation process **550** is completed, an operator may begin providing proppant to in the fluid provided to the fracturing pumps **200** by activating the blender screw conveyors **412** (Step **580**) and/or may manually change the discharge rate of the fracturing pumps **200** (Step **582**). When the operator takes control at Steps **580**, **582**, the operator may reactivate an automatic or scheduled program of the operation the controller **30** returns to monitoring the fluid pressure increase rate of Step **530**.

(27) Returning back to the entry into the mitigation process **550**, when the fluid pressure increase rate is a Tier 3 Potential Screen Out and the fluid pressure is below or less than the first preselected percentage of the maximum fluid pressure, e.g., 90%, the controller **30** maintains the discharge rate of the fracturing pumps **200** and the delivery of the proppant (Step **556**). When the discharge rate of the fracturing pumps **200** and the delivery of the proppant is maintained, an operator may provide input to the controller **30** to manually change the discharge rate of the fracturing pumps **200** or reactivate an automatic or scheduled program to the operation of the controller **30** (Step **582**). If an operator does not intervene, the controller **30** continues to monitor fluid pressure.

(28) If the operator does not intervene and the fluid pressure reaches a second preselected percentage of the maximum fluid pressure, e.g., 94%, the controller **30** intervenes by preparing for and running the process **560**. Specifically, the controller **30** prepares for the process **560** by stopping the blender screw conveyors **412** to stop delivery of proppant (Step **554**). When the delivery of proppant is stopped, the controller **30** begins the process **560** to incrementally decrease a discharge rate of the fracturing pumps **200** as detailed above until by cycling through Step **562** and Step **564** until the fluid pressure increase rate is no longer increasing or stabilized, e.g., less than or equal to zero. When the fluid pressure increase rate is stabilized, the discharge rate of the fracturing pumps **200** is maintained (Step **570**) such that the mitigation process **550** is complete or terminated. When the mitigation process **550** is completed, an operator may begin providing proppant to in the fluid provided to the fracturing pumps **200** by activating the blender screw conveyors **412** (Step **580**) and/or may manually change the discharge rate of the fracturing pumps **200** (Step **582**). When the operator takes control at Steps **580**, **582**, the operator may reactivate an automatic or scheduled program of the operation the controller **30** returns to monitoring the fluid pressure increase rate of Step **530**.

(29) The mitigation process **550** enables the controller **30** to automatically stop delivery of proppant to the fluid provided to the fracturing pumps **200** and to decrease the discharge rate of the fracturing pumps **200** until the fluid pressure increase rate is stabilized without input from an operator. During the mitigation process **550**, including the process **560**, an operator may be prevented or locked out from certain commands of the controller **30**. For example, in some embodiments, during the mitigation process **550**, an operator may be locked out of all commands to the controller **30** except at step **556** until the mitigation process **550** such that the fluid pressure increase rate has been stabilized. In certain embodiments, an operator may be locked out of increasing the discharge rate of the fracturing pumps **200** or initiating or increasing delivery of proppant during the mitigation process **550**.

(30) By reducing well screen out, the need for operations to reopen fractures or a well (e.g., wire line operations) may be reduced

or eliminated such that time, and thus costs, to stimulate a well may be reduced. In addition, the method **500** of detecting and mitigating well screen out with a controller **30** may reduce rapid pressure increases associated with well screen outs such that stress on fracturing equipment may be reduced. The fracturing equipment may include, but not be limited to, fracturing pumps, power end assemblies of power units (e.g., gas turbine engines), gearboxes, transmissions, and piping or iron of the well site. Further, by intervening before the fluid supplied to the wellhead reaches the maximum fluid pressure, reliance on pressure relief valves, such as a wellhead pressure relief valve, may be reduced. Reducing reliance on pressure relief valves may conserve energy by not releasing pressure within the system and reduce stress on the fracturing equipment by maintaining a more consistent fluid pressure within the maximum wellhead pressure.

(31) The method **500** being executed by the controller **30** allows for continuous monitoring of the fluid pressure and the fluid pressure increase rate at higher rate (e.g., 1 Hz to 300 Hz) when compared to relying on manual control and monitoring. In addition, by including multiple tiers of warnings (e.g., Tier 1 and Tier 2) the controller **30** alerts an operator to intervene before the fluid pressure approaches the maximum wellhead pressure and may automatically intervene if the fluid pressure increase rate reaches Tier 3 and the fluid pressure approaches the maximum wellhead pressure.

(32) This is a continuation of U.S. Non-Provisional application Ser. No. 17/991,007, filed Nov. 21, 2022, titled “METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT,” now U.S. Pat. No. 11,939,854 issued Mar. 26, 2024, which is a continuation of U.S. Non-Provisional application Ser. No. 17/355,920, filed Jun. 23, 2021, titled “METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT,” now U.S. Pat. No. 11,566,506 issued Jan. 31, 2023, which is continuation of U.S. Non-Provisional application Ser. No. 17/303,841, filed Jun. 9, 2021, titled “METHODS AND SYSTEMS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT,” now U.S. Pat. No. 11,208,881, issued Dec. 28, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 17/182,408, filed Feb. 23, 2021, titled “METHODS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT,” now U.S. Pat. No. 11,066,915, issued Jul. 20, 2021, which claims priority to and the benefit of U.S. Provisional Application No. 62/705,050, filed Jun. 9, 2020, titled “METHODS AND SYSTEMS FOR DETECTION AND MITIGATION OF WELL SCREEN OUT,” the disclosures of which are incorporated herein by reference in their entireties.

(33) The foregoing description of the disclosure illustrates and describes various exemplary embodiments. Various additions, modifications, changes, etc., may be made to the exemplary embodiments without departing from the spirit and scope of the disclosure. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense. Additionally, the disclosure shows and describes only selected embodiments of the disclosure, but the disclosure is capable of use in various other combinations, modifications, and environments and is capable of changes or modifications within the scope of the inventive concept as expressed herein, commensurate with the above teachings, and/or within the skill or knowledge of the relevant art. Furthermore, certain features and characteristics of each embodiment may be selectively interchanged and applied to other illustrated and non-illustrated embodiments of the disclosure.

Claims

1. A wellsite hydraulic fracturing pumper system, the system comprising: one or more fracturing pumps configured to pump fluid to a wellhead when positioned at a hydrocarbon well site; one or more pressure transducers positioned at a location of one or more of: (a) adjacent an output of the one or more fracturing pumps, and (b) at the wellhead, the one or more pressure transducers each configured to measure a fluid pressure of the fluid provided to the wellhead; and a controller to control the one or more fracturing pumps and positioned in signal communication with the one or more pressure transducers such that the controller receives the fluid pressure of the fluid provided to the wellhead, the controller (a) including memory, a processor to process data, and a screen out detection and mitigation protocol program stored in the memory and (b) being responsive to a process in which the protocol of the controller incrementally decreases a discharge rate of the one or more fracturing pumps, in response to: (i) a fluid pressure increase rate of the fluid supplied to the wellhead being greater than a preselected increase rate, and (ii) the fluid pressure of the fluid provided to the wellhead being greater than a preselected percentage of a maximum wellhead pressure, until the fluid pressure is stabilized.

2. The wellsite hydraulic fracturing pumper system according to claim 1, wherein the screen out detection and mitigation protocol includes an alarm to provide an alert indicative of when the fluid pressure increase rate is greater than the preselected increase rate before the pressure of the fluid provided to the wellhead is within the preselected percentage of the maximum wellhead pressure.

3. The wellsite hydraulic fracturing pumper system according to claim 2, wherein the one or more pressure transducers also are positioned at a location adjacent the output of the one or more fracturing pumps.

4. The wellsite hydraulic fracturing pumper system according to claim 3, further comprising one or more blenders configured to provide fluid and proppant to the one or more fracturing pumps, wherein each of the one or more blenders includes a blender conveyor configured to rotate such that proppant is provided to the one or more fracturing pumps in response to rotation of the blender conveyor.

5. The wellsite hydraulic fracturing pumper system according to claim 4, the controller further being responsive to a process in which the protocol of the controller incrementally decreases a flow rate of the one or more blenders, wherein incrementally decreasing the discharge rate of the one or more fracturing pumps and the flow rate of the one or more blenders by the controller includes stopping delivery of proppant at the one or more blenders prior to decreasing the discharge rate of the fracturing pump when the one or more blenders are delivering proppant.

6. The wellsite hydraulic fracturing pumper system according to claim 5, wherein incrementally decreasing the discharge rate of the one or more fracturing pumps and the flow rate of the one or more blenders by the controller includes decreasing the discharge rate of the one or more fracturing pumps prior to decreasing the flow rate of the one or more blenders.

7. A wellsite hydraulic fracturing pumper system, the system comprising: one or more fracturing pumps configured to pump fluid to a wellhead when positioned at a hydrocarbon well site; one or more pressure transducers positioned at a location of one or more of: (a) adjacent an output of the one or more fracturing pumps, (b) at the wellhead, the one or more pressure transducers each configured to measure a fluid pressure of the fluid provided to the wellhead, and (c) adjacent the output of the one or more

fracturing pumps; and a controller to control the one or more fracturing pumps and positioned in signal communication with the one or more pressure transducers such that the controller receives the fluid pressure of the fluid provided to the wellhead, the controller configured to incrementally decrease a discharge rate of the one or more fracturing pumps in response to: (i) a fluid pressure increase rate of the fluid supplied to the wellhead being greater than a preselected increase rate, and (ii) the fluid pressure of the fluid provided to the wellhead being greater than a preselected percentage of a maximum wellhead pressure, until the fluid pressure is stabilized.

8. The wellsite hydraulic fracturing pumper system according to claim 7, wherein the one or more pressure transducers also are positioned at a location adjacent the output of the one or more fracturing pumps.

9. The wellsite hydraulic fracturing pumper system according to claim 7, further comprising one or more blenders configured to provide fluid and proppant to the one or more fracturing pumps.

10. The wellsite hydraulic fracturing pumper system according to claim 9, the controller further being responsive to a process in which the protocol of the controller incrementally decreases a flow rate of the one or more blenders, wherein incrementally decreasing the discharge rate of the one or more fracturing pumps and the flow rate of the one or more blenders by the controller includes stopping delivery of proppant at the one or more blenders prior to decreasing the discharge rate of the fracturing pump when the one or more blenders are delivering proppant.

11. The wellsite hydraulic fracturing pumper system according to claim 10, wherein incrementally decreasing the discharge rate of the one or more fracturing pumps and the flow rate of the one or more blenders by the controller includes decreasing the discharge rate of the one or more fracturing pumps prior to decreasing the flow rate of the one or more blenders.

12. The wellsite hydraulic fracturing pumper system according to claim 7, wherein the controller is configured to incrementally decrease the discharge rate of the one or more fracturing pumps by one or more increments, each increment in the range of 0.5 barrels per minute to 10 barrels per minute.

13. A wellsite hydraulic fracturing pumper system, the system comprising: one or more mobile fracturing pump units to pump fluid to a wellhead when positioned at a hydrocarbon well site, each of the fracturing pump units comprising a gas turbine engine configured to drive a fracturing pump; one or more pressure transducers configured to be positioned at a location of one or more of: (a) adjacent an output of the one or more fracturing pumps, or (b) at the wellhead, and to measure a fluid pressure of the fluid when provided to the wellhead; and a controller to control the one or more fracturing pumps and configured to be positioned in signal communication with the one or more pressure transducers such that the controller receives measurement of the fluid pressure of the fluid when provided to the wellhead and controllably stabilizes the fluid pressure when the fluid pressure supplied to the wellhead increases greater than a preselected increase rate and being greater than a preselected percentage of a maximum wellhead pressure, the controller configured to incrementally decrease a discharge rate of the one or more fracturing pumps, in response to a fluid pressure measurement obtained by the one or more pressure transducers.

14. The wellsite hydraulic fracturing pumper system according to claim 13, wherein the controller comprises a memory, a processor to process data, and a screen out detection and mitigation protocol program stored in the memory, the controller being responsive to a process in which the protocol of the controller incrementally decreases a discharge rate of the one or more fracturing pumps, in response to: (i) a fluid pressure increase rate of the fluid supplied to the wellhead being greater than a preselected increase rate, and (ii) the fluid pressure of the fluid provided to the wellhead being greater than a preselected percentage of a maximum wellhead pressure.

15. The wellsite hydraulic fracturing pumper system according to claim 14, wherein the screen out detection and mitigation protocol includes an alarm to provide an alert indicative of when the fluid pressure increase rate is greater than the preselected increase rate before the pressure of the fluid provided to the wellhead is within the preselected percentage of the maximum wellhead pressure.

16. The wellsite hydraulic fracturing pumper system according to claim 15, wherein the controller is configured to incrementally decrease the discharge rate of the one or more fracturing pumps by one or more increments, each increment in the range of 0.5 barrels per minute to 10 barrels per minute.

17. The wellsite hydraulic fracturing pumper system according to claim 16, wherein the controller is configured to delay determination of a fluid pressure increase rate after incrementally decreasing the discharge rate of the one or more fracturing pumps by the one or more increments.

18. The wellsite hydraulic fracturing pumper system according to claim 13, wherein the one or more pressure transducers also are positioned at a location adjacent the output of the one or more fracturing pumps.

19. The wellsite hydraulic fracturing pumper system according to claim 13, wherein the controller is configured to incrementally decrease the discharge rate of the one or more fracturing pumps by one or more increments, each increment in the range of 0.5 barrels per minute to 10 barrels per minute.
